二元一次方程组

代入法　消元法（高斯消元法）

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 & ① \\ a_{21}x_1 + a_{22}x_2 = b_2 & ② \end{cases}$$

① × a_{22} − ② × a_{12}

$\therefore (a_{11}a_{22} - a_{21}a_{12})x_1 = b_1a_{22} - b_2a_{12}$

$\therefore x_1 = \dfrac{b_1a_{22} - b_2a_{12}}{a_{11}a_{22} - a_{21}a_{12}}$　同理 $x_2 = \dfrac{a_{11}b_2 - a_{21}b_1}{a_{11}a_{22} - a_{12}a_{21}}$　(*)

引入行列式

$D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$　2×2　行　列

副对角线　主对角线

*：a_{11} a_{12} a_{21} a_{22} 的差。

$D = a_{11}a_{22} - a_{12}a_{21} = \begin{vmatrix} a_{11} & a_{12} \\ a_{22} & a_{21} \end{vmatrix}$

*式：$x_1 = \dfrac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}$　$x_2 = \dfrac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}$

*：分母是系数

分子则把常数项取代需求的那列系数

克拉默法则

三元一次方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \end{cases}$$

$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \neq 0$

则 $x_1 = \dfrac{D_1}{D}$　$x_2 = \dfrac{D_2}{D}$　$x_3 = \dfrac{D_3}{D}$

*：D_1 为常数项取代系一列

为 $\begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}$　其它同理.